

Date:13-07-2026

Name:-Nagasai

Reg No:-192424292

1.Mapping color

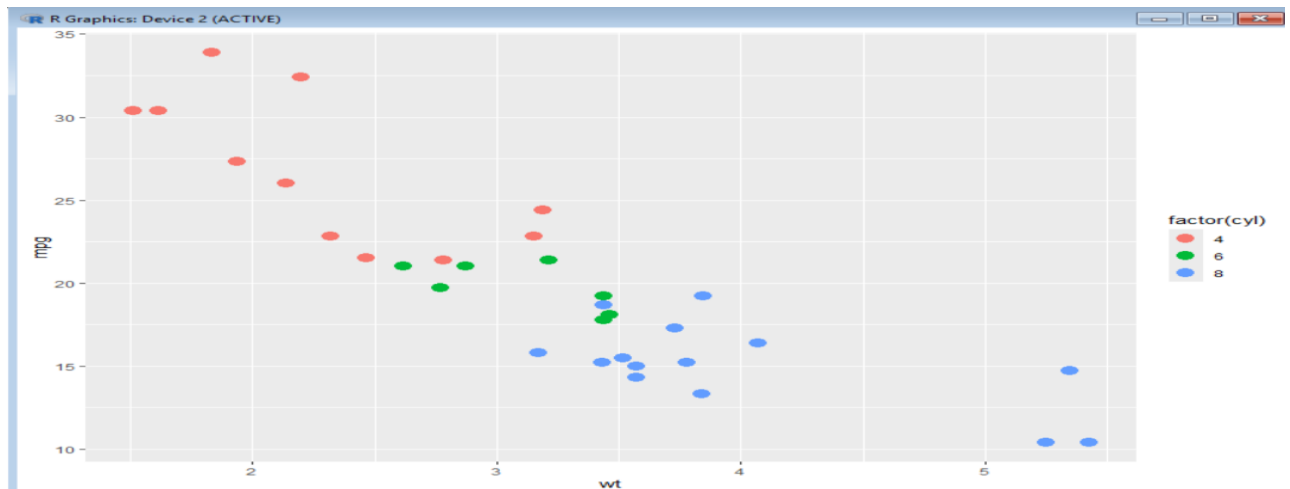
Code:

```
library(ggplot2)

ggplot(mtcars, aes(x = wt, y = mpg, color = factor(cyl))) +

  geom_point(size = 3)
```

output:



2.Continuous example

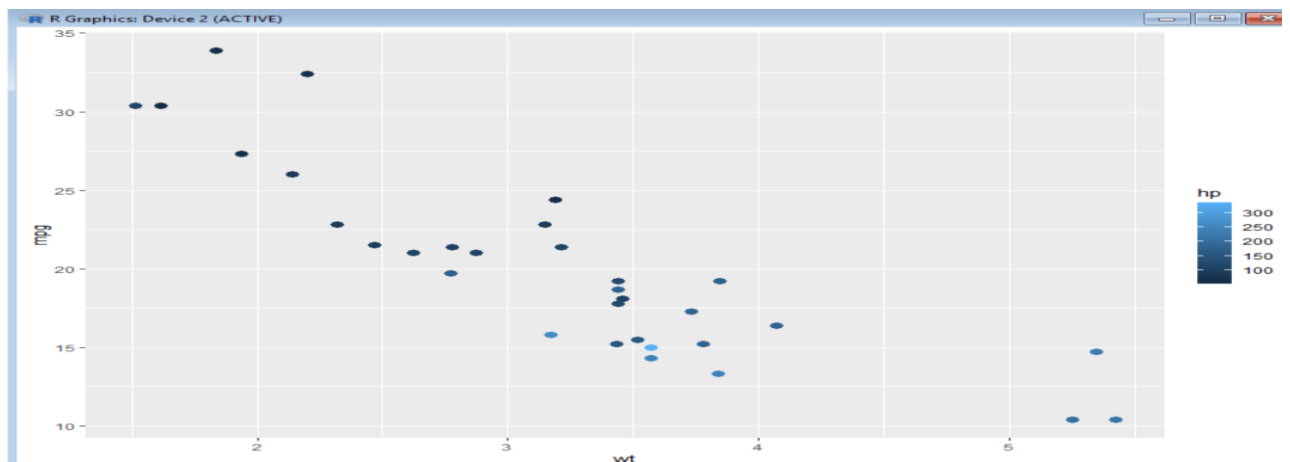
Code:

```
library(ggplot2)

ggplot(mtcars, aes(x = wt, y = mpg, color = hp)) +

  geom_point(size = 3)
```

output:



3.Categorical example

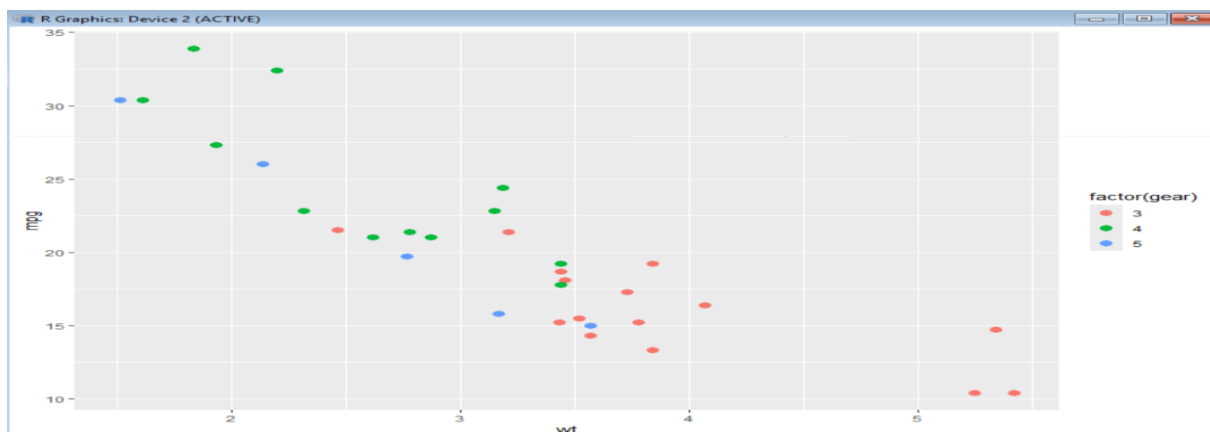
Code:

```
library(ggplot2)

ggplot(mtcars, aes(wt, mpg, color = factor(gear))) +

  geom_point(size = 3)
```

output:



4.Modify color scale

Code:

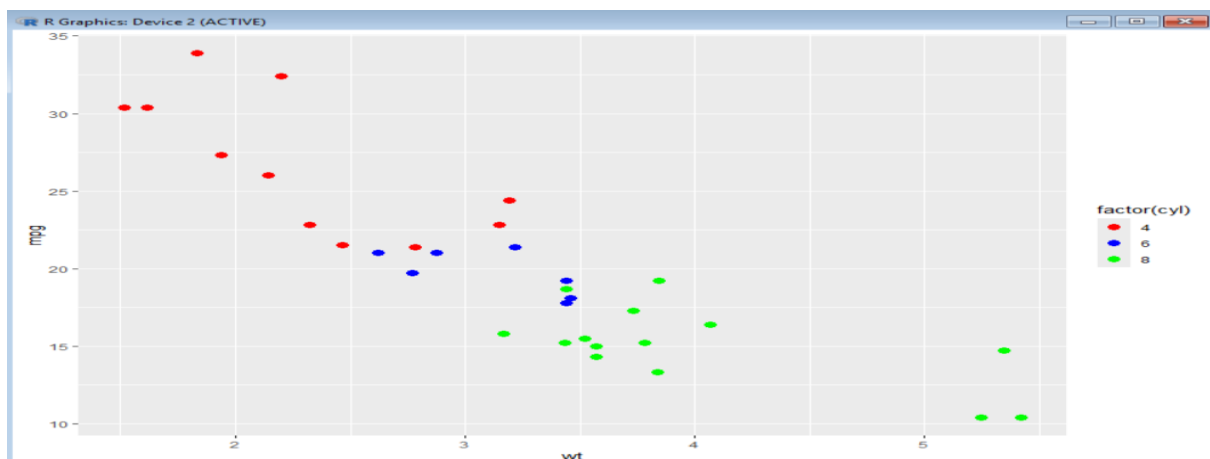
```
library(ggplot2)

ggplot(mtcars, aes(wt, mpg, color = factor(cyl))) +

  geom_point(size = 3) +

  scale_color_manual(values = c("red", "blue", "green"))
```

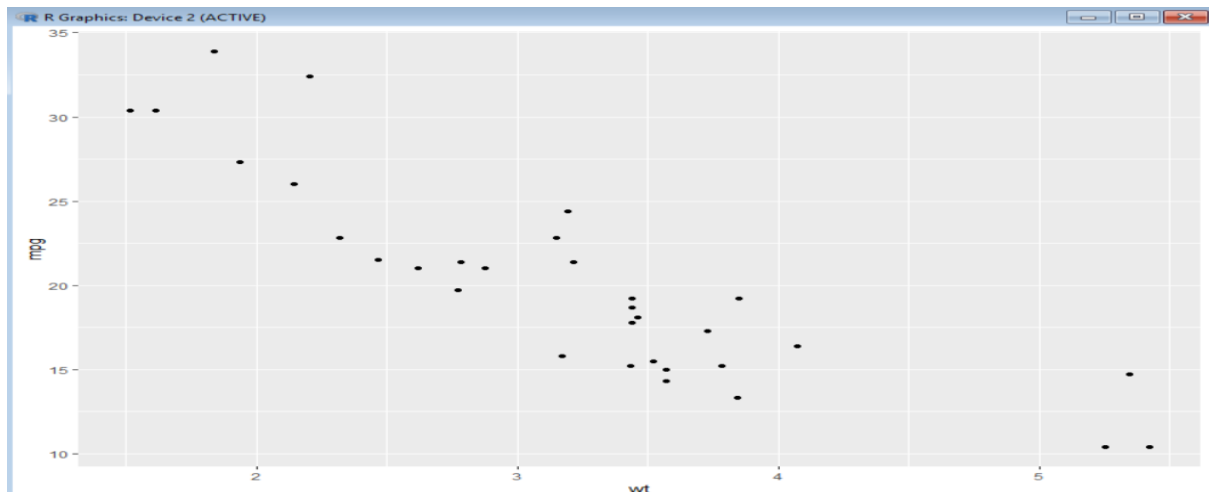
output:



5. Custom breaks

Code:

```
library(ggplot2)
ggplot(mtcars, aes(wt, mpg)) +
  geom_point() +
  scale_x_continuous(breaks = seq(1, 6, 1))
```

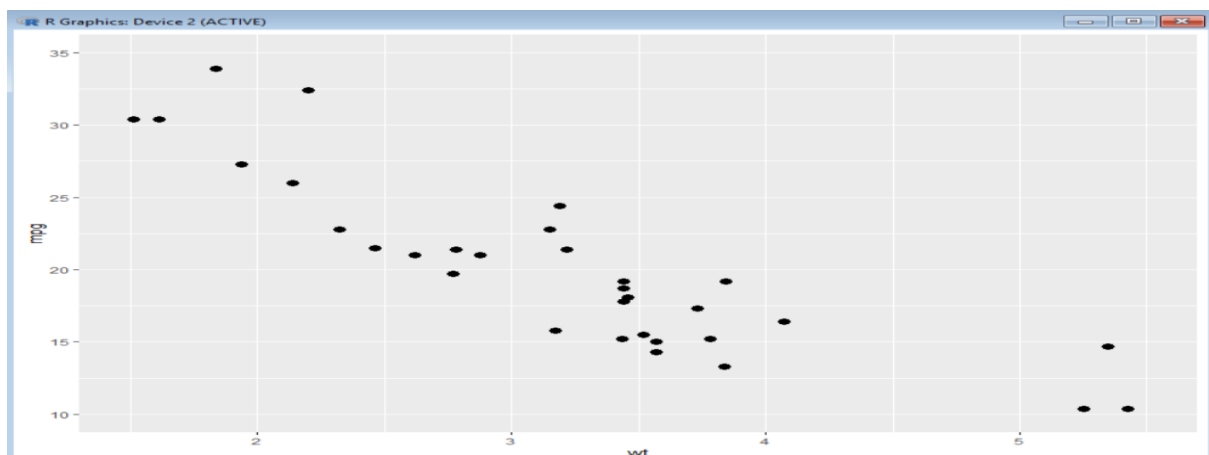


6. Cartesian coordinates

Code:

```
library(ggplot2)
ggplot(mtcars, aes(wt, mpg)) +
  geom_point(size = 3) +
  coord_cartesian(xlim = c(1.5, 5.5), ylim = c(10, 35))
```

output:

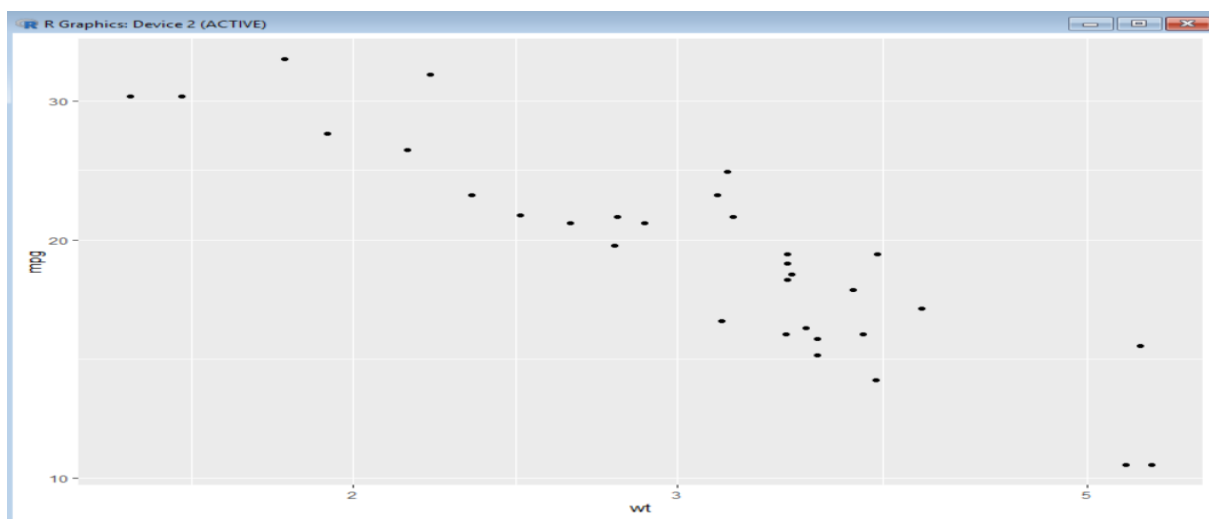


7.Non linear axes

Code:

```
library(ggplot2)
ggplot(mtcars, aes(wt, mpg)) +
  geom_point() +
  scale_x_log10() +
  scale_y_log10()
```

output:

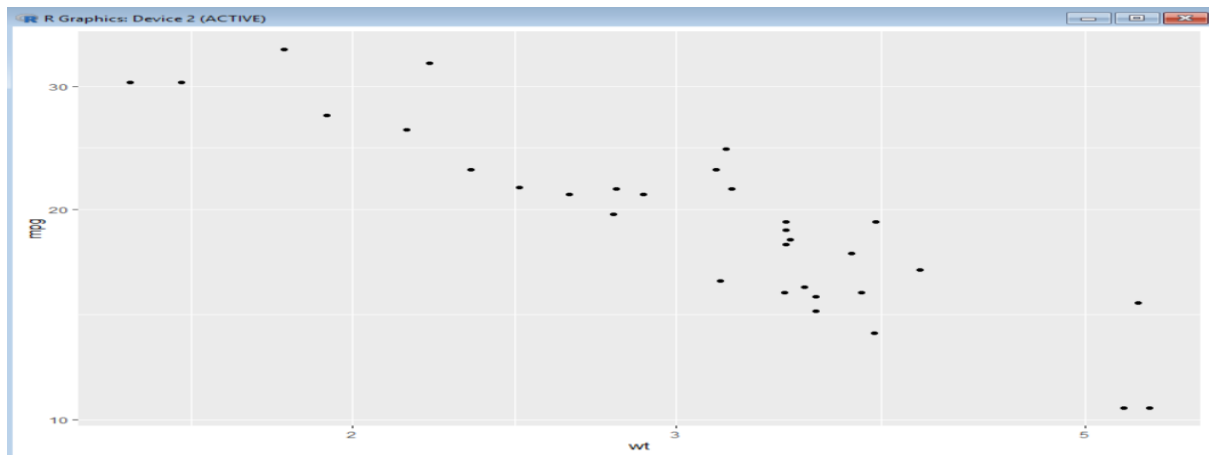


8.log scale example

Code:

```
library(ggplot2)
ggplot(mtcars, aes(wt, mpg)) +
  geom_point() +
  scale_x_log10() +
  scale_y_log10()
```

output:

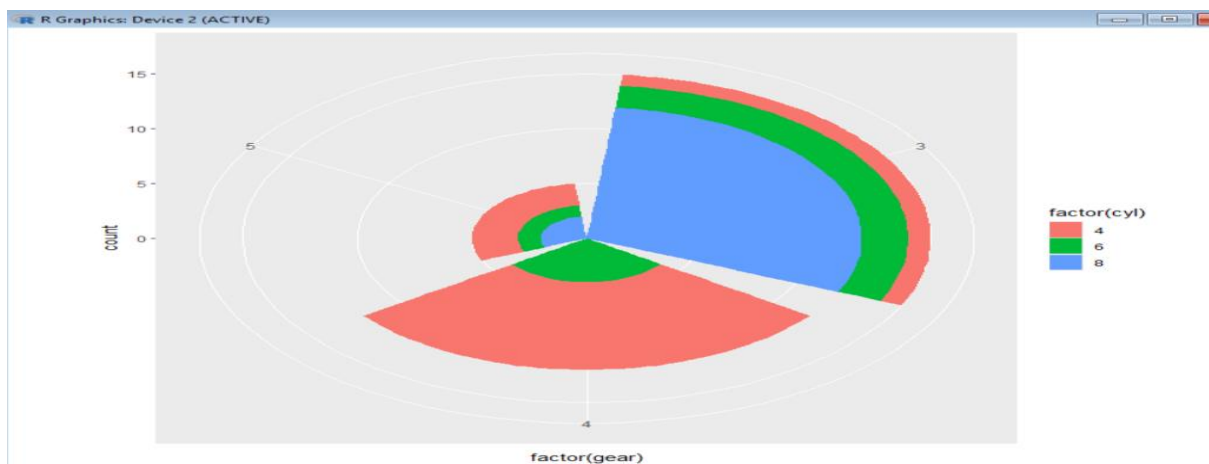


9.polar coordinates

Code:

```
library(ggplot2)
ggplot(mtcars, aes(x = factor(gear), fill = factor(cyl))) +
  geom_bar() +
  coord_polar()
```

output:

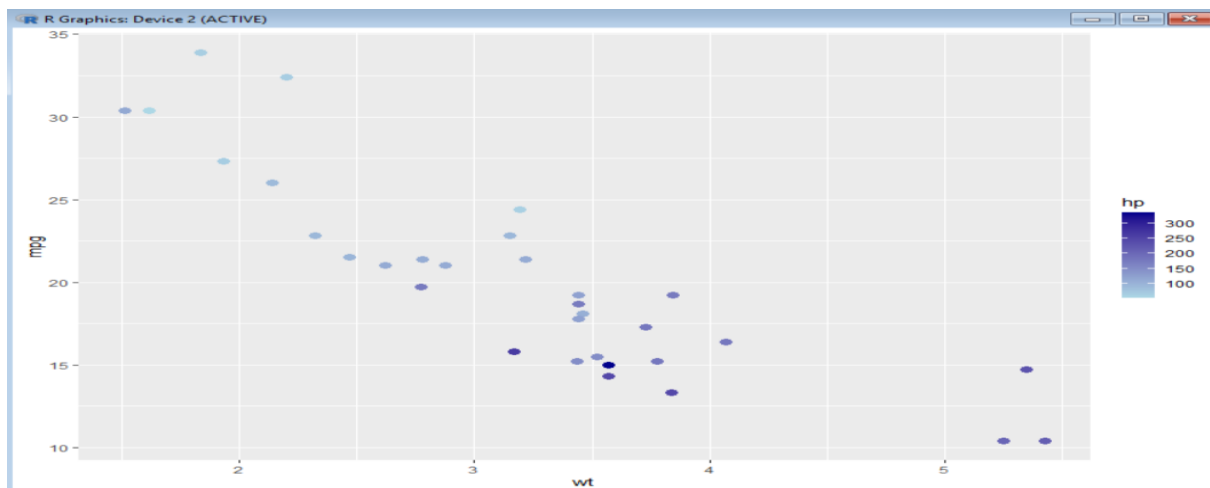


10.sequential color

Code:

```
library(ggplot2)
ggplot(mtcars, aes(wt, mpg, color = hp)) +
  geom_point(size = 3) +
  scale_color_gradient(low = "lightblue", high = "darkblue")
```

output:

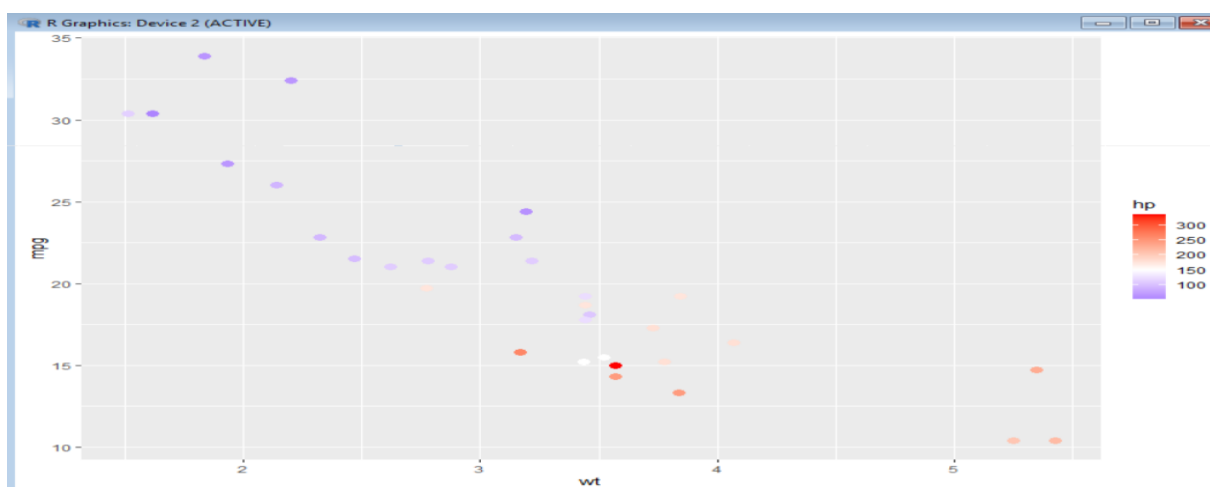


11.Diverging color

Code:

```
library(ggplot2)
ggplot(mtcars, aes(wt, mpg, color = hp)) +
  geom_point(size = 3) +
  scale_color_gradient2(low = "blue",
                        mid = "white",
                        high = "red",
                        midpoint = 150)
```

output:



12.Color as a tool distinguish

Code:

```
library(ggplot2)
```

```
ggplot(mtcars, aes(wt, mpg, color = factor(cyl))) +
```

```
  geom_point(size = 3)
```

output:

